Azza Sawungrana

Head of Data at Geosquare

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PROFESSIONAL SUMMARY

Head of Data at Geosquare, where national datasets become location intelligence for land, hazard, property and mobility decisions. I own the platform architecture, the standards the team works to, and what it costs to run.

The thread through Bhumi Varta and Geosquare has been the same one: taking delivery that was built for one client and making it work for the next. My work sits where data engineering meets product judgement, and I care about the cost of serving the second client, not only the quality of the first map.

EXPERIENCE

Head of Data

Geosquare, Jakarta, Indonesia (remote) — Aug 2025 - Present

A location intelligence company building grid-based data products for land, property, risk and network decisions.

- Lead the data function: platform architecture, engineering standards, the team, and what it costs to run.
- Took a platform built for one national land agency and sold it to two more clients, with a third in discussion. The first client is now on its third phase.
- Cut total cloud spend by 32%. The whole national dataset now sits in object storage for about $23 a month, so a new client adds compute rather than another copy of the data.
- Moved the bus factor from 2 people who could run the pipeline end to end to 5.
- Wrote the platform and portfolio strategy: six reusable engines, six product lines, and the sequence for getting from one to the other.
- Building KYL, a recurring product priced by the area a client needs.

Geospatial Analyst

Van Lanschot Kempen, Amsterdam, Netherlands — May 2025 - Jul 2025

Internship with the SDG Farmland Fund, on how a global farmland portfolio reports on sustainability and climate risk.

- Evaluated the fund's existing data systems and defined the use cases worth building first.
- Wrote the conceptual design for geospatial analysis, visualisation and reporting across the portfolio.

Lead Product Manager

Bhumi Varta Technology, Tangerang, Indonesia — Apr 2020 - Feb 2023

LOKASI Intelligence, a location intelligence platform used by Indonesian banks, retailers and FMCG companies to decide where to expand and how to run their networks.

- Shipped version 2 in April 2021, a full redesign and refactor of the v1 codebase.
- Led LOKASI Intelligence Enterprise and turned client-specific deployments into modular capability inside the main product, instead of one-off forks.
- Ran the product team for most of this period with no CPO above it, so strategy, roadmap and prioritisation were mine.
- Built the delivery process for a small team: requirement gathering, PRD standards, agile, and a DevOps setup that worked without a dedicated ops person.

Product Consultant

Bhumi Varta Technology, Remote — Feb 2023 - Jun 2023

Handover role before moving to the Netherlands.

Founder and Head of Product

Nusadventure, Klaten, Indonesia — Feb 2018 - May 2019

A sustainable tourism venture built with Desa Wisata Jarum.

- Built and ran the product, operations and content side.
- Closed it after two years when conversion could not support the team.

Geospatial Data Analyst

Coordinating Ministry for Economic Affairs, Jakarta, Indonesia — Mar 2017 - Jan 2018

One Map Policy, the national effort to reconcile conflicting land designations.

- Built and integrated thematic permit maps (mining, cultivation rights, location permits) for provinces across Sumatra and Nusa Tenggara.
- Worked with regional planning agencies, land offices and mining departments to trace overlapping designations back to their source.

EDUCATION

MSc Spatial Engineering

University of Twente, ITC, Enschede, Netherlands — Sep 2023 - Jul 2025

A project-based programme about solving societal problems with spatial data. Funded by the ITC4SE scholarship.

- Thesis: migrant children's well-being and urban green and blue spaces in Enschede.
- Projects: drought and qanat water management in Mashhad; green-blue-grey space mapping for Kampala; local EV charging for a congested grid in Zwolle.

BSc Cartography and Remote Sensing

Universitas Gadjah Mada, Yogyakarta, Indonesia — Sep 2012 - Feb 2017

Thesis on lahar (volcanic mudflow) modelling for Kelud, using field samples from three municipalities.

- Laboratory assistant for photogrammetry, disaster mapping and non-photographic remote sensing.
- Led the Geography Study Club for a year.

PUBLICATIONS

- Geosquare System v1 - a nation-centric hierarchical grid. Chapter in a UGM Press volume edited by Trias Aditya. In press.
- Geosquare System v2 - country datums and per-country coverages. arXiv preprint. In preparation.
- Age-cohort dasymetric allocation to 50 m cells (RIUH). FIT ISI 2026, Semarang. Presented; IOP proceedings in preparation.

SKILLS

- **Data engineering:** Python, DuckDB, PostGIS, GDAL, GeoParquet, PMTiles, Cloud optimised GeoTIFF, object storage, cloud cost control
- **Geospatial:** remote sensing, discrete global grid systems, dasymetric mapping, isochrones and routing, multi-criteria suitability, hazard modelling, spatial validation
- **Methods:** Ridge, random forest, gradient boosting and kernel models, spatial and temporal holdouts, uncertainty reporting, reproducible pipelines
- **Leadership:** hiring, performance management, technical writing and standards, product strategy, cross-disciplinary delivery

LANGUAGES

- Indonesian — Native
- Javanese — Native
- English — Professional working proficiency